

**Self-Verifying Axiom Systems
and the Incompleteness Theorem**

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Self-Verifying Axiom Systems and the Incompleteness Theorem

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Abstract: We introduce some weak First Order axiom systems which can *simultaneously* verify their own consistency and prove more Π_1 theorems than Peano Arithmetic. The rule of inference used by these axiom systems may be semantic tableaux, or any other cut-free deduction method. These systems will not violate Godel's Incompleteness Theorem essentially because they will treat non-zero division, *rather than multiplication*, as a total function. The definition of self-consistency employed in this paper may be either knowledge of the non-existence of a proof of $0=1$ or a strong version of the Σ_1 reflection principle. Our systems can be modified to support Gentzen-style deduction with cuts, if Subtraction and Division replace Addition and Multiplications as the declared total functions. We also prove that these results do not generalize when Π_1 reflection is present, and show that certain versions of our axiom system become inconsistent when multiplication replaces division as one of the formally defined atomic functions.

Most of the material in this paper is quite technical, but Sections 1 thru 4 were written so that they should be comprehensible to a skilled researcher with a more modest background in Logic.

NOTE TO THE READER: This paper is not quite as long as its *paper weight* might suggest. The formal theorem proving is only 20 pages long (pp. 13-32). The remainder consists mostly of an informal 12-page intuitive introduction and a 16-page description of the formal Godel encoding of IS(A) (in Appendix A). The latter Godel encoding was inserted into this article only for the sake of completeness: A reader can fully understand the gist without delving into Appendix A.

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1. Introduction: Define an axiom system A to be self-verifying iff

- i) one of the theorems implied by A is the statement of its own consistency, and
- ii) the axiom system is in fact consistent.

Rogers has noted that Kleene's Fixed Point Theorem implies that every recursively enumerable axiom system A can be easily expanded into a broader system A^* which satisfies condition (i) [Kl38,Ro67]. The catch is that A^* can be inconsistent even while it asserts its own consistency, hence violating condition (ii). This problem arises not only in Godel's paradigm, where A has at least the strength of Peano Arithmetic (PA), but also for many axiom systems which are much weaker than PA. For instance, it is known that even weakened axiom systems that include only PA's Σ_1^0 – Induction Principle can not be self-verifying [Fe88,Sm83], and [Bu86] provides proofs of some versions of Godel's Second Theorem for certain very weak cut-free systems.

Examples of self-verifying axiom systems in the prior literature appear in [Bo79, Je71, Fe60, Kr71, KT74, Ro67, Ta53]. It is difficult to make exact comparisons between our proposed IS(A) system and any of the previous self-verifying systems because Godel's Incompleteness Theorem requires that every self verifying system must breach in some way one of the three fundamental Hilbert-Bernays conditions [Me87]. The breach, which can be very subtle, may arise because of either the axiom system employed or the choice of deduction method, but not necessarily due to both. For instance, [Fe60] illustrates a self-verifying system which employs all the axioms of Peano Arithmetic, but which draws upon a deduction method which captures the numerical but not intensional definition of classic deduction. The IS(A) formalism, studied by us, will be a class of First Order self-verifying systems which can prove more Π_1 theorems than Peano Arithmetic and simultaneously rely upon a deduction method as natural as semantic tableaux. However to preclude IS(A) from becoming inconsistent, several of the axioms of Peano Arithmetic, including the axiom that the multiplication function is total, are weakened under IS(A). [Ta53,KT74] have illustrated some fascinating Second Order cut-free self-verifying systems which can prove all the translated mappings of the theorems of Peano Arithmetic into a second order language.

Robert Solovay [So94] recently proved a new version of Godel's Incompleteness Theorem, shortly after learning of our work. Solovay's theorem implies that it is impossible for an axiom system, satisfying both parts of the definition of self verification, to prove all Peano Arithmetic's Π_1 theorems about the Subtraction and Non-zero Division functions while *simultaneously*:

- a) employing a Gentzen-style deduction method **WITH A CUT RULE**, and
- b) recognizing Addition (or even successor) as a total function.

Solovay's Theorem is the almost direct complement to our Propositions 1 & 7, in that these theorems show that self-verifying axioms systems can be constructed when *either* condition (a) or (b) is relaxed.

Propositions 2 thru 6 will further tighten our results by clarifying

- 1) what types of reflection principles are and are not compatible with self verification.
- 2) in what sense is a total multiplication function incompatible with a cut-free self-verifying axiom system.

Every self-verifying axiom systems will be of course subject to very tight limits upon its permissible strength, by the limits imposed by the Godel Incompleteness Theorem. From this perspective, Feferman draws the distinction between an intensionally and numerically correct definition of deduction, Solovay's new theorem imposes strict limits on self-verifying systems using Gentzen-style deductive cuts, and it is well known that no self-verifying formalism can satisfy the three Hilbert-Bernays conditions [HB39,Me87].

But yet on the other hand, human beings appear to implicitly presume (*at some type of weak level ?*) their own consistency when they reason because otherwise it is awkward (*if not impossible ?*) to justify the mental energy expended during cogitation. Thus, it is a valid research topic to investigate how powerful, in the light of Godel's Theorem, can self-verifying systems become ?

2. General Perspectives:

This section will define the general *Introspective Semantics* (IS), which shall be capable of proving their own consistency. Let A denote a recursive axiom system. Then the axiom system IS(A) will consist essentially of the following three groups of axioms.

- 1) an initial axiom group defining a preliminary set of functions and relations,
- 2) the statement that all Π_1 theorems (defined formally on the next page), proven by "the inner axiom system" A, are true;
- 3) the statement that the full system IS(A), including this third sentence, is consistent. (Several alternate means for formalizing this "*self-verification assertion*" will be explored.)

Our chief goal will be to assure that if A is consistent then it will automatically follow that IS(A) is also consistent. As noted in section 1, there are many examples of well-defined axiom systems

which assert their own consistency but which are inconsistent. An example is PA^* which consists of all the axioms of Peano Arithmetic (PA) plus the assertion "*Peano Arithmetic is consistent, and so is the combination of Peano Arithmetic with this last statement*" [Ro67]. Our main theorem will show that IS(A) does not suffer an inconsistency problem analogous to PA^* .

Point of Clarification: Say $f(x_1, x_2, \dots, x_n)$ is a **non-growth function** iff it satisfies $f(x_1, x_2, \dots, x_n) \leq \text{Max}(x_1, x_2, \dots, x_n)$, and call $f(x_1, x_2, \dots, x_n)$ a **growth function** otherwise. All our formalisms will employ non-growth functions, such as Subtraction and non-zero Division. The difficulty in proving IS(A) satisfies part (ii) of the definition of "self-verification" will increase dramatically as one extends IS(A)'s Group-1 Axioms to include more growth functions. Thus, our systems will fall into three categories depending on whether they have no growth functions, permit Addition as a function, or recognize both Multiplication and Addition as total functions. It will be only the first category of self-verifying systems, analyzed in Proposition 7, that will be capable of permitting Gentzen-style deductive cuts without becoming inconsistent. Solovay's Theorem will imply that these systems can not be extended to include even a Successor function. Propositions 1-4 will show that there are reasonable examples of self-verifying systems that recognize Addition as a function, provided cut-free deduction is employed. Proposition 6 will show that there are examples of cut-free IS(A) systems becoming inconsistent when Multiplication is added to their Group-1 functions!

Added Introductory Comment: The results, mentioned above, are quite unlike Presburger Arithmetic because IS(A) will recognize non-zero Division as a total function (and hence can view Multiplication as a well-defined partial function).

We will now define IS(A) formally. Since it will be desirable to view subtraction and division *as total functions* over the set of non-negative integers, $x - y$ is defined to equal zero when $x \leq y$, and $x / 0$ is defined to equal zero. In five of our seven main theorems, the symbol G_0 will designate IS(A)'s set of Group-1 axioms. The eight formally defined functions of G_0 will consist of Addition, Subtraction, Division, plus the following further functions:

- i) **StringCount** (x, y, j) = the number of repetitions of the j -bit binary encoding of the integer y in x when one views x as a sequence of j -bit alphabet symbols of length $= \lceil \log_2(x+1) / j \rceil$.
- ii) **Shift** (x, y) designating the integer $z = x / 2^y$.
- iii) **Extract** (x, y, z) designating the integer w which stores the same values as x between bit-positions y and z , and stores zeroes everywhere else.
- iv) **Andreverse** (x, y, i, j) designating the string obtained by reversing x 's sequence of bits between the addresses i and j and then taking the bitwise AND of the result with y .
- v) **Address** (x, y, z, i, j) indicating the least $k \leq \lceil \log_2 x \rceil / j$ where if s is a string enumerating x 's first kj bits then $\text{StringCount}(s, y, j) - \text{StringCount}(s, z, j) = i$ (if such k exists), and $\text{Address}(x, y, z, i, j) = 0$ otherwise

Other axioms in G_0 will include the definitions of the usual relations equality and greater-than, the statement $\forall x : x+1 \neq x$, and for each integer constant $\bar{k} = \bar{1}, \bar{2}, \bar{3}, \dots$, the two special axioms $\bar{k} = \overline{k-1} + \bar{1}$ and $\forall x : x < \bar{k} \supset x = \bar{0} \vee x = \bar{1} \vee \dots \vee x = \overline{k-1}$.

Notation: An existential or universal quantifier will be called **bounded** iff it occurs in a form $\exists v \leq z$ or $\forall v \leq z$, where z denotes either a constant or variable symbol. A wff $\Phi(x_1 x_2 \dots x_j)$, whose only free variables are $x_1 x_2 \dots x_j$ will be called a Δ_0 formula iff all its universal and existential quantifiers are bounded. Given such a Δ_0 formula Φ and $i \leq j$, the canonical form of a Π_1 and Σ_1 formula is defined as $\forall x_1 \forall x_2 \dots \forall x_i : \Phi(x_1 x_2 \dots x_j)$ and $\exists x_1 \exists x_2 \dots \exists x_i : \Phi(x_1 x_2 \dots x_j)$. A sentence will be defined as a formula with no free variables, and the Δ_0 , Σ_1 , Π_1 and Π_2 sentences will thus be defined accordingly.

Formal Definition of the Axiom Group G_0 : The axiom indicating that the addition function is total can obviously be written as a Π_2 sentence. Since all G_0 's other functions are non-growth functions, its remaining axioms can be written as Π_1 sentences. Any finite set of Π_1 axioms, validly defining G_0 's functions and relations, is adequate.

Define a first-order axiom system A to be *nice* iff A is consistent with the Group-1 axioms of IS and there exists a Δ_0 formula $Ax_A(y)$ expressing that y is the Godel number of an axiom of A . For technical reasons, it is convenient to view all encodings of proofs as one of semantic tableaux, resolution, or cut-free sequent calculus. The Godel Completeness Theorem [Go30] holds for each of these methods [Be56,Fi90,Ge69,Sc56,Sm68,Ta87]. For simplicity, we rely mostly on the semantic tableaux proofs. $Prf_A(x,y)$ will thus denote the Δ_0 formula asserting that y is a semantic tableaux proof of x from a nice system A . Since $Turing(a,b,c)$ and $Ax_A(d)$ are both Δ_0 formulae, it will be possible to encode $Prf_A(x,y)$ as a Δ_0 formula, as Section 5 and its Appendix will explain.

In our discussion, $Pr_A(x)$ denotes the formula " $\exists y Prf_A(x,y)$ ", and $[\Phi]$ denotes Φ 's Godel number. For each Π_1 sentence Φ , the second axiom group will contain one sentence of the form:

$$Pr_A([\Phi]) \supset \Phi \quad (2.1)$$

Note (2.1) is Π_1 (when written in prenex normal form) because $Prf_A(x,y)$ is Δ_0

In its simplest form, the third axiom group of $IS(A)$ consists of a single sentence asserting "The union of the first two groups of axioms *with this very sentence* forms a consistent set of axioms." This self-referential sentence is known to be well defined via fixed point operators (see [Ro67,Je71]). (We will formally define it in the Appendix to Section 5.) As noted before, all proofs in this paper are conducted in the semantic tableaux formalism [Fi90,Sm68]. In this

formalism, an inconsistency proof is a semantic tableaux whose axioms are those of $IS(A)$, where all paths of the semantic tableaux are closed. If $\perp$ is the empty sentence (or equivalently the statement $0 = 1$) then self-consistency can be formalized as the statement:

$$\forall y \neg \text{Prf}_{IS(A)}(\perp, y) \quad (2.2)$$

The first theorem of this paper, stated formally in the next section, asserts that if A is nice then $IS(A)$ must be consistent. This paper will contain six other theorems. Our result would not be very strong if (2.2) was the sole respect in which this paper established self-consistency (because *in the special context of weak cut-free axiom systems* assertion (2.2) is not ideally informative). Our formal results are listed in the next section. The remainder of this section will explain intuitively why IS treats addition as a function, but multiplication only as a relation.

The axiom system IS was intended to enable computerized algorithms to prove their own consistency. For simplicity, let us temporarily assume our formal system is a finite state machine with b bits of memory. Then the formal system will be technically unable to construct integers larger than 2^b . Thus in one sense, assertion (2.3) can be viewed as *incorrect* because z can exceed 2^b when x and y are b -bit integers exceeding 2^{b-1} .

$$\forall x \forall y \exists z : z = x + y \quad (2.3)$$

Nevertheless, there will be another very different, somewhat *informal interpretation* of Equation (2.3), where the expression can be viewed as valid, even when the variables x , y and z have fixed bounded domains. Let us interpret (2.3) as stating whenever our finite state machine has enough storage space to ask the question "What is the value of $x + y$?", it also has enough memory available to store the answer. In particular, since the symbol "+" must require at least one bit of storage, the variables x and y must require fewer than $b-1$ bits, implying there is sufficient space in a b -bit memory to store their sum z .

The analog of (2.3) is, surprisingly, inapplicable to multiplication. Consider the equation

$$\forall x \exists z : z = x * x \quad (2.4)$$

Then if $x = 2^{3b/4}$, the right hand side of (2.4) can be stored in fewer than b bits of memory while $z > 2^b$ can not be! At first, it may appear that this counterexample is artificial because it requires at least two occurrences of the same variable x on the right sides of (2.4) to force $z > 2^b$. However, the proof of Godel's Incompleteness Theorem used essentially the same double appearance of a variable when it provided a counterexample to self-verification via a diagonalization argument. One of our

theorems will show that the classic paradigm, where an axiom system becomes automatically inconsistent as soon as it proves its own consistency, is inapplicable to the cut-free versions of IS.

The axiom family IS(A), can be intuitively regarded as a class of structures, lying halfway between Peano Arithmetic and Presburger Arithmetic, which satisfy Godel's First but not Second Incompleteness Theorem. That is, since the predicate Turing (x,y,z) is expressible under IS(A), there clearly can exist no decision procedure for classifying the theorems provable from IS(A). On the other hand, unlike Peano Arithmetic, the various versions of IS(A) will be able to verify their own consistency.

Moreover *from at least some perspectives*, IS does not lose much information relevant to pragmatic computation by viewing multiplication as *not being* a function. Consider the example

$$\Phi(w) = \{ \forall x < \sqrt{w} \ \forall y < \sqrt{w} \ \exists z < w : z = x * y \} \quad (2.5)$$

Then IS(Peano) can prove the sentence $\forall w \Phi(w)$ (by using the Group-2 axioms and the fact that this sentence is provable in Peano Arithmetic). This example and its many generalizations illustrate that although IS(Peano) can not prove multiplication formally to be a function, it will still be able to treat multiplication as a function in a wide class of concrete applications involving integers (see *).

Or to be more precise, the preceding paragraph and footnote *illustrate at least one reasonable criteria for examining the pragmatic*. They are not meant to imply that other equally reasonable criteria would not find IS(Peano) to be lacking. To the contrary, in a context where the Incompleteness Theorem indicates that *every* axiom system *either fails* to prove its consistency or fails to meet the full Hilbert-Bernays conditions, this would not be a reasonable expectation.

Another way to motivate the axiom system IS is to consider a philosopher's response to the question of whether multiplication is a function. His first answer would likely be "*of course yes*". But if one then asked the philosopher to construct a sequence of natural numbers $a_0, a_1, a_2, a_3 \dots$ where $a_0 \geq 2$ and $a_{i+1} = (a_i)^2$, a hedge would quickly follow because a_n requires at least 2^n bits. The expected response would then be that "*multiplication is a function in theory but not always in practice*". Note that the philosopher would not give such a quick answer about addition because the sequence $b_0, b_1, b_2, b_3 \dots$ with $b_{i+1} = b_i + b_i$ is characterized by proofs of the existence of b_n

* The point is that concrete applications (or at least *very concrete applications*) can usually be encoded so that some type of approximate analog of w bounds all the pertaining multiplicative products z when one encodes the problem in a suitably reasonable form. It is extremely rare for a pragmatic application to involve numbers so large that some type of analog of w is *unavailable for all possible natural encodings of the problem* ! (Also, it should be noted that IS can represent floating point multiplication as a *total function*, provided some form of constraints are present on the bit-length of the mantissa.)

requiring more bits than b_n 's binary encoding (in cut-free proof systems where the Principle of Induction is unavailable). This means there will always be adequate memory for writing b_n 's binary encoding when there is sufficient memory for storing a semantic tableaux proof of its existence (whereas the same is plainly not true *when multiplication is assumed to be a function*).

For instance, multiplication allows one to establish the existence of a number a_{300} (in a proof with approximately 300 lines), whose binary encoding requires *more digits* than the number of atoms in the universe. In sharp contrast, even the full cardinality of the universe is *insufficient* to construct a cut-free proof from the Addition Axioms of the existence of a number whose *bit length* is as large as the universe. (The caveats that the Principle of Induction is absent and that cut-free deduction is employed are necessary for the preceding metaphor to retain meaning.)

The seven main theorems of this paper will generalize this metaphor by showing how the fast growth properties of the multiplication function prevents cut-free axiom systems from being self-verifying. As an over-simplified metaphor, our IS(A) system shall circumvent these difficulties by adopting the perspective of the philosopher who declares "*multiplication is a function in theory but not always in practice*". Thus, the inner axiom system A of IS(A) corresponds to the theoretical aspect, which can view multiplication as a function. On the other hand, the outer three groups of IS(A)'s axioms stopped just short of recognizing multiplication as a function, so as to preclude a "fast-growing" sequence $a_0, a_1, a_2, a_3 \dots$ forcing IS(A) to become inconsistent.

The last paragraph called the preceding examples "*over-simplified metaphors*" because they are applicable only to cut-free systems. These metaphors lose meaning when cuts are present because Addition can then be employed to *implicitly define* a sequence that grows actually faster than $a_0, a_1, a_2, a_3 \dots$. Solovay's proof that IS(A) is incompatible with deductive cuts [So94] rests on this fast growth property. (Actually, IS(A) can support a cut rule for Δ_0 sentences, and Solovay's theorem relies on the cut rule for sentences with unbounded quantifiers.)

Finally, it should be reiterated that the preceding metaphors were intended only to help motivate the subject matter. They offer an accurate description of cut-free systems, but are over-simplistic when cuts are present. The true direction of this article will not be apparent until after the next chapter lists and explains the significance of our seven main theorems. By studying IS(A), we seek to learn *to what precise extent can one formalize the notion of self-consistency*. Both positive and negative results will be derived, in regards to which deductive methods, which reflection principles, and what type of growth functions IS(A) can and cannot support. It should be stressed that Equation (2.2), in isolation, is a fairly weak form of the assertion of self-consistency. The seven

theorems, listed in the next section, offer a much better summary of our results.

3. Formal Summary of Main Results:

Section 6 of this paper will prove:

Proposition 1: For each nice axiom system A , $IS(A)$ is consistent.

A disadvantage to weak axiom systems, such as $IS(A)$, is that several conventional definitions of consistency, which are equivalent under Peano Arithmetic, are not also equivalent under $IS(A)$. Thus, most of this paper will attempt to generalize Proposition 1 for stronger notions of self-verification than was indicated by Equation (2.2).

Some added notation is needed. The symbol $PA+$ will denote the natural generalization of Peano Arithmetic that includes the eight atomic functions from IS 's Group-1 axioms. The symbol $IS_d^s(A, G)$ will denote the generalization of $IS(A)$ where d is the deduction method employed, G is an arbitrary set of Group-1 axioms and s is some type of Group-3 class providing some type of self-verification statement. Whenever we omit one or more of the symbols d , G or s from $IS_d^s(A, G)$, they will assume the default definitions given in Section 2.

The local reflection statement for a sentence Φ and *fixed* axiom system B is the assertion:

$$Pr_B(\lceil \Phi \rceil) \supset \Phi \quad (3.1)$$

Some articles omit the word "local" when referring to reflection principles.

We will also study four stronger (and more "uniform") versions of the reflection principle. Let $\Phi(v)$ denote a formula with only v free, and Φ_i denote the sentence $\Phi(i)$. Then Eq. (3.2) thru (3.4) will be called B 's *first uniform*, *second uniform* and *semi-uniform* reflection principles for Φ

$$\{\forall v Pr_B(\lceil \Phi_v \rceil)\} \supset \forall v \Phi(v) \quad (3.2)$$

$$\forall v \{Pr_B(\lceil \Phi_v \rceil) \supset \Phi(v)\} \quad (3.3)$$

$$Pr_B(\lceil \forall v \Phi(v) \rceil) \supset \Phi_i \text{ for all fixed integers } i \quad (3.4)$$

Define an existential quantifier to be *bounded* by y iff it is an expression similar to $\exists u < y$. If Φ_v is a Σ_1 sentence, define Φ_v^y to be a modified form of Φ_v which requires all its existential quantifiers to be bounded by y . The *strong-uniform* Σ_1 reflection principle for an axiom system B is the statement that all Σ_1 formulae $\Phi(v)$ satisfy

$$\forall v \forall y \{Prf_B(\lceil \Phi_v \rceil, y) \supset \Phi_v^y\} \quad (3.5)$$

Lob's Theorem states that traditional axiom systems are unable to prove *any* of their reflection principles, except for in the degenerate case where the right side of the implicant is provable by them. See [Bo79] for an extremely simple proof of Lob's Theorem [Lo55, Me87], and [Fe62, Sm83] for a discussion of when the first and second uniform principles are equivalent.

Proposition 2 thru 4 (below) illustrate that introspective axiom systems support reflection properties more fully than traditional axiom systems. These theorems are important because the reflection principles represent a much stronger form of self-verification than Equation (2.2). (See footnote * for an intuitive explanation.)

Proposition 2. (proven in Section 7) In contrast to the conventional axiom systems discussed by Lob's Theorem, $IS(PA+)$ can prove

- a) its local reflection statements for *every* Δ_0 sentence,
- b) its local reflection statement for the subset of Π_1 sentences Ψ which are decidable,
- c) its semi-uniform reflection statement *whenever* Φ_i in Equation (3.4) is Δ_0 .

Proposition 3. (proven in Section 8) Define $IS^{\Sigma_1}(A)$ as an axiom system whose Group-1 and Group-2 axioms are identical to those of $IS(A)$ and where Group-3 now includes the stronger assertion that $IS^{\Sigma_1}(A)$'s strong-uniform reflection property is valid for all Σ_1 formulae. *Then for any nice A , $IS^{\Sigma_1}(A)$ is consistent.*

Proposition 4. (proven in Section 9) The system $IS^{\Sigma_1}(PA+)$ (as well as $IS^{L\Sigma_1}(PA+)$ and $IS^{U\Sigma_1}(PA+)$ defined in Remark 1) *can prove Φ 's semi-uniform reflection statements when Φ_i is Σ_1 .*

Remark 1. Define $IS^{L\Delta_0}(A)$, $IS^{L\Sigma_1}(A)$, $IS^{U\Sigma_1}(A)$ and $IS^{U\Delta_0}(A)$ to be four alternate axiom systems whose Group 3 axioms instead assert their local Δ_0 reflection, local Σ_1 reflection, second uniform Σ_1 reflection, and second uniform Δ_0 reflection properties, respectively. Since each of these systems are weaker than $IS^{\Sigma_1}(A)$, Proposition 3 immediately implies their consistencies. Also, Propositions 2 & 4 generalize for any A which is an extension of $PA+$.

Proposition 5 and 6 (below) are negative results. They illustrate respects in which each of Propositions 2, 3 and 4 can not be strengthened. In particular, they show that the addition of either Π_1 reflection or the multiplication function to the preceding systems will produce inconsistencies!

* Equation (2.2) merely indicates that $IS(A)$ knows it can not prove $\perp$. This does not automatically guarantee that upon proving $\neg\Phi$, $IS(A)$ can automatically infer that it can *not* prove Φ . It needs access to some reflection principle, such as for example Equation (3.1) with $B = IS(A)$, to conclude the latter from the proof of $\neg\Phi$!

Proposition 5. (proven in Section 11) Let $IS^{L\Pi_1}(A)$ denote an axiom system whose Group-1 and Group-2 axioms are again identical to those of $IS(A)$, but where Group-3 now asserts that the $IS^{L\Pi_1}(A)$'s Local Reflection Property is valid for all Π_1 sentences. *Then there exists nice A where $IS^{L\Pi_1}(A)$ is inconsistent.* (An example of a nice system forcing introspective inconsistencies in Propositions 5 & 6 is $PA+$, or any extension of it.)

Proposition 6. (proven in Section 12) Let G_m denote an extension of G_0 which contains the added axiom that the multiplication function is total. Recall that Remark 1 indicated that $IS^{U\Delta_0}(A) = IS^{U\Delta_0}(A, G_0)$ was consistent for all nice A. *Nevertheless, there do exist nice A where $IS^{U\Delta_0}(A, G_m)$ is inconsistent.* (This result obviously applies also to $IS^{U\Sigma_1}(A, G_m)$ and $IS^{\Sigma_1}(A, G_m)$, since the latter two axiom systems are extensions of $IS^{U\Delta_0}(A, G_m)$.)

The most surprising facet of the preceding results is the contrast between Proposition 5, which asserts that introspective axiom systems are incompatible with even a weak form of Π_1 reflection, and the quite close approximations to Π_1 reflection established in Propositions 2c and 4. (See also Section 9 for the more general partial reflexive implications of proofs of Π_n theorems.)

In all six of the preceding theorems, the subscript d in $IS_d^*(A, G)$ was omitted, indicating the deduction corresponded by default to semantic tableaux. It is trivial to generalize Propositions 1-6 for the cases where d corresponds to resolution [Fi90] and the cut-free sequent calculus [Ta87], since these methods exclude unnecessary uses of the Law of the Excluded Middle in a manner analogous to semantic tableaux. The subscript + will henceforth denote deduction systems that permit cuts, such as Hilbert systems. (See [Ta87] for a formal definition of cuts.) Our main theorem about self-verifying $IS_+(A)$ systems appears below:

Proposition 7. Let G_- denote a set of Group-1 axioms identical to G_0 except that G_- does not include the axiom indicating that Addition is a total function. *Then for any nice A, $IS_+(A, G_-)$ is consistent.* (Generalizations of Proposition 7 and its proof appear in Section 13.)

As noted in Section 1, Robert Solovay [So94] discovered a new version of Godel's Second Incompleteness Theorem, shortly after learning of our results. It is stated below:

Solovay's Theorem. There exists a finite set F of the Π_1 theorems proven in Peano Arithmetic such that no consistent Gentzen-style sequent calculus system employing cuts can simultaneously prove all the theorems of F, recognize Subtraction, non-zero Division and Successor as total functions, and prove its own consistency.

Let G_s denote a set of Group-1 axioms identical to G_0 and G_- (in Propositions 1 & 7), except that Successor is the only growth function recognized by G_s . Then Solovay's Theorem implies that $IS_+(PA+, G_s)$ is inconsistent. This fact is quite startling because Solovay's inconsistent system $IS_+(PA+, G_s)$ would be transformed into a consistent system if *either*

- a) deductive cuts were removed from the formalism (i.e. apply Proposition 1),
- b) or if the successor function was removed from G_s (i.e. apply Proposition 7).

Thus, Solovay's Theorem and the other negative results in Propositions 5 & 6 define three different variants of inconsistent axiom systems, whose formal definitions differ only slightly from the well-behaved systems in Propositions 1-4 and 7. (Actually, the fit between Solovay's Theorem and Proposition 1 is even tighter than indicated in item (a) because 1) $IS(PA+)$ can support a cut rule for Δ_0 sentences, and 2) $IS(PA+)$ recognizes Addition as a total function while $IS_+(PA+)$ can not tolerate even Successor !)

One final definition will help further clarify the significance of Propositions 1-4 & 7. Say a formal system S satisfies the Δ_0 cut-reflection principle iff for any pair of Δ_0 sentences α and β , S can prove

$$\forall x \forall y : \{ \text{Prf}_S(\ulcorner \alpha \urcorner, x) \wedge \text{Prf}_S(\ulcorner \alpha \supset \beta \urcorner, y) \} \supset \beta \quad (3.6)$$

Say S satisfies Equation (3.6) in the strong sense iff S 's semantic tableaux proof of (3.6) has a structure independent of α and β (this means that for any pair of Δ_0 sentences (α', β') , the proof of (3.6) for (α', β') will be the same as the proof for (α, β) except that each appearance of the sentences α and β in the first proof will be replaced by α' and β' .) Lastly, say that S satisfies the strong uniform Δ_0 cut-reflection principle iff the preceding conditions are satisfied, and also for any pair of Δ_0 formulae $\Phi(a)$ and $\Psi(b)$, S can prove the following *uniformized generalization* of Equation (3.6) :

$$\forall x \forall y \forall a \forall b : \{ \text{Prf}_S(\ulcorner \Phi_a \urcorner, x) \wedge \text{Prf}_S(\ulcorner \Phi_a \supset \Phi_b \urcorner, y) \} \supset \Phi_b \quad (3.7)$$

A trivial corollary to Proposition 2a is that $IS(PA+)$ satisfies the weak form of the Δ_0 cut-reflection principle. The proof of this corollary is only two or three sentences long, and it is equally easy to verify that the systems $IS^{L\Delta_0}(PA+)$ and $IS^{U\Delta_0}(PA+)$ (from Remark 1) satisfy the strong and strong-uniform versions of the Δ_0 cut-reflection principle, respectively. This result, in turn, implies that Proposition 3's $IS^{\Sigma_1}(PA+)$ system satisfies the strong-uniform version of the Δ_0 cut-

reflection principle because $IS^{\Sigma_1}(PA+)$ is strictly stronger than $IS^{U\Delta_0}(PA+)$.

The results above are surprising because classical axiom systems are incapable of proving any version of the Δ_0 cut-reflection principles (see *). Thus while Solovay's Theorem indicates that nearly all our $IS(A)$ systems are incapable of accommodating a Gentzen-style cut rule for eliminating sentences with unbounded quantifiers, there is nevertheless a different type of Δ_0 cut principle which is compatible with our *but not with conventional logic formalisms !!!!*

A well-known corollary to the Godel Incompleteness Theorem is that every self-verifying system must violate one of the Hilbert-Bernays conditions [HB39,Me87]. For the case of $IS(A)$, the difficulty is that only in the *degenerate case* where x and y are *fixed constants* can $IS(A)$ prove the Hilbert-Bernays condition that *if x is a proof of α and y is a proof of $\alpha \supset \beta$ then there exists some z that proves β* . This weakness of $IS(A)$, together with the other limitations on it imposed by Solovay's Theorem and Propositions 5 & 6, clearly illustrate how $IS(A)$ can not fully characterize the complexity of human thought. But on the other hand, it would also appear to be simplistic to fully ignore self-verifying systems because humans appear to implicitly presume their consistency (at some type ? of weak level), for the sake of justifying the energy expended during cogitation.

We suggest that the truth (?) may possibly lie between these two extremes. That both the self-verifying and non-self-verifying perspectives may convey some useful partial insights. Such a pluralistic view is partially analogous to Godel's First Incompleteness Theorem, which indicates that no single r.e. axiom system can enumerate all the theorems one desires to learn about arithmetic.

Or at least in a context where a modest change in a formalism's definition will cause Propositions 1 thru 7 and Solovay's Theorem to produce starkly different results, it appears premature to develop hasty conclusions, whether positive or negative, about self-verifying systems.

One reason we suspect that it would be premature to dismiss self-verifying axiom systems in an overly hasty manner is that the preceding discussion has shown how some self-verifying systems can support Equation (3.7)'s strong uniform Δ_0 cut-reflection principle, while Peano Arithmetic and all other conventional axiom systems are incompatible with it.

* Consider the example where $\alpha \equiv \beta \equiv \{0=1\}$. A conventional axiom system would be incapable of proving (3.6) for such α and β because it would then verify its own consistency !!

4. Intuition Behind the Proof of Proposition 1

The formal proof of Proposition 1 appears in Section 6. In this section, we will sketch the intuition behind the proof. The proof of Proposition 1 will be by contradiction. It will assume that p is the minimal integer which is a Godel encoding of an inconsistency proof using the axiom system $IS(A)$, and show this assumption leads to a contradiction.

The proof utilizes the fact that if A is nice, then all its Group-1 and Group-2 axioms are consistent with each other. Thus p can represent an inconsistency proof only if the final axiom sentence in Group-3 contradicts the union of the Group-1 and Group-2 axioms.

The single sentence in Group-3 was the statement that " $IS(A)$ is consistent". Section 6 will prove that the Group-1 and 2 axioms can contradict this statement only if the proof p constructs *within itself* a second "witness integer" p^* (not necessarily distinct from p) such that p^* is also a proof of $IS(A)$'s inconsistency.

Since p is defined to be the minimal integer encoding a contradiction proof, it follows that the witness p^* satisfies $p^* \geq p$. If $IS(A)$ had included multiplication as a function, it would be possible for an inconsistency proof p to construct within itself a representation of the necessary $p^* \geq p$. (Essentially, section 2 noted that in strictly fewer than $\log_2 p$ bits, it is possible to encode via multiplication numbers $p^* \geq p$. The construction of p^* "inside p " in the proof of Proposition 6 will employ essentially the the analog of the Eq. (2.4) implied by Godel's Incompleteness Theorem.) However a crucial difference between Propositions 1 and 6 is that the Group-1 Axioms of the former contain only Addition as a growth function. This restriction makes impossible the construction of the contradiction witness p^* "inside p ". In particular, Addition grows so slowly that $\log(x + y) \leq \text{MAX}(\log x, \log y) + 1$. This implies essentially that it is impossible in $\log_2 p - 1$ bits to encode a number as large as p . Using these facts, Section 6 will prove that p cannot possibly be long enough to construct the necessary witness $p^* \geq p$ for p to be an inconsistency proof !!

Added Remark. For the sake of simplicity, we have mentioned only Propositions 1 & 6 during the intuitive discussion above. However, similar reasoning is necessary to prove Propositions 3, 5, 7 and Solovay's Theorem. That is, the positive self-verification results in Propositions 3 & 7 rest on showing that it is impossible to construct the necessary contradiction witness $p^* \geq p$ "inside p ". Similarly, the negative inconsistency results indicated in Proposition 5 and Solovay's Theorem depend ultimately on showing how to construct the contrary witness $p^* \geq p$ "inside p ".

5. Semantic Tableaux Formalism.

The formal proofs of Propositions 1-7 appear in Sections 6-13 (essentially one proof per section). The first version of our proofs was conceived using Gentzen's cut-free sequent calculus notation. Later, we observed that our proofs could be simplified significantly, if we instead employed the semantic tableaux notation [Fi90,Sm68], sometimes also called semantic trees, because it mixes exceptionally well with Section 6's notation. This section will outline the formal encodings of semantic tableaux proofs and related preliminary information.

One difficulty will be that multiplication (and hence also x^λ (for *fixed exponent* λ)) are not functions under IS. Lemma 5.1 illustrates how we can overcome this difficulty by treating x^λ *almost* as if it was a function!!!

Lemma 5.1. Say a formula $\Phi(x_1x_2...x_L)$ is majorized with exponent λ iff each universally and existentially quantified variable v in Φ is required to be less than x_i^λ for some $i < L$. Then for each majorized formula $\Psi(x_1x_2...x_L)$ there exists an "essentially equivalent" Δ_0 formula $\Psi^*(x_1x_2...x_L)$ such that every L -tuple $(i_1, i_2...i_L)$ satisfies $\Psi(i_1, i_2...i_L) \equiv \Psi^*(i_1, i_2...i_L)$.

Informal Proof Sketch. To construct Ψ^* we will replace each $\exists v < x^\lambda$ with the sequence $\exists v_1 < x \exists v_2 < x \cdots \exists v_{2\lambda} < x$, and treat universal quantifiers analogously. Define Ψ^* to be the natural modification of Ψ when the string v is so divided into 2λ corresponding smaller strings. (The string v is divided into 2λ rather than λ substrings to avoid round-off errors.) It is obvious that such a construction will produce a Ψ^* satisfying Lemma 5.1. QED

Let E1 and E2 denote two different methodologies for encoding a proof formalism. Define E1 and E2 to be **linearly equivalent** iff each proof string in E1 can be translated into a string in E2 (and vice versa), where the translations may increase the lengths of the encodings by no more than a constant factor. Lemma 5.1 implies that the set of Δ_0 formulae for two distinct but linearly equivalent encodings must be essentially isomorphic. (In the context of conventional axiom systems, where multiplication is treated as a function, one typically ignores such minute levels of detail. Lemma 5.1 assures that the absence of a formal multiplication function in IS(A) does not raise any new complications, with regards to linearly equivalent encodings.)

Define a byte to be an unit consisting of six bits. Our encoding of wffs and proofs will comprise a string of bytes whose length is essentially guaranteed to be proportional to the length of

the English language sentence making the same statement (so as to achieve the ideal "linear" bit compression). In particular, we will assign special 6-bit codes to the following symbols:

- 1) the standard six logical connectives: $\wedge$, $\vee$, $\neg$, $\supset$, $\forall$, and $\exists$
- 2) three types of left and right parenthesis symbols: (,), [,], {, and }, and a comma symbol.
- 3) five conventional symbols $\hat{A}$, $\hat{v}$, $\hat{c}$, $\hat{u}$ and $\hat{f}$ to denote an atomic formula, a variable, a fixed constant, a proof-parameter and a function in a semantic tableaux proof [Fi90].
- 4) two special new symbols $u\#$ and $v\#$, explained in the next paragraph, which allows us to omit repeatedly storing long constants. Instead, the symbols $u\#$ and $v\#$ allow us to use the "u" and "v" symbols to act as pointers to constants.

Integers i will be encoded using $\lfloor \log_{32} i \rfloor + 3$ bytes (i.e. one byte to initiate the integer code, $\lfloor \log_{32} i \rfloor + 1$ bytes to encode i as a base-32 number, and a final byte serving as a termination symbol). The byte strings $\hat{u}[i]$ and $\hat{v}[i]$, representing the i -th parameter and variable, will be thus encoded using $\lfloor \log_{32} i \rfloor + 3$ bytes. Similarly, the i -th function (with j arguments $t_1 t_2 \cdots t_j$) will be encoded as $\hat{f}[i,j] [t_1, t_2 \cdots t_j]$, and $\hat{A}[i,j] [t_1, t_2 \cdots t_j]$ will encode the i -th atomic formula.

The symbol $\hat{c}[j]$ will also have a quite standard meaning, it will designate a constant that represents the integer j . If j is a very large number, then its repeated appearance in a proof could cause the proof to grow to an unacceptably large size. Therefore, our notation will employ two special symbols to reduce the bit length of a proof. The symbol $u\#[i, j]$ may appear in any position normally occupied by the constant symbol $\hat{c}[j]$. It will treat the indicated sentence as if $\hat{c}[j]$ was stored in $u\#[i,j]$'s position, and also specify that *all appearances* of $\hat{u}[i]$ in the subtree descending from this sentence should be treated *as if it was the constant $\hat{c}[j]$* (written only in an abbreviated form). Similarly the sentence " $v\#[i, j] \Phi (\hat{v}[i])$ " is a bit compressed representation for $\Phi(\hat{c}[j])$. Part of the reason for this new notation is explained by Lemma 5.2.

Lemma 5.2. For fixed j and variable Φ , $IS(PA+)$ can prove that there exists a function that maps the Godel number of $\lceil \Phi(\hat{v}[1]) \rceil$ onto the Godel number for $\lceil v\#[1, j] \Phi (\hat{v}[1]) \rceil$. Also, it recognizes a function that maps the byte string " $\hat{c}[j]$ " onto $\lceil v\#[1, j] \Phi (\hat{v}[1]) \rceil$ when j is a variable and Φ is fixed. (The reason for the new " $v\#$ " notation is that the two analogous mappings onto $\lceil \Phi(\hat{c}[j]) \rceil$ are unknown by $IS(PA+)$ to be total functions !)

Proof of Lemma 5.2's First Sentence. Since PA^+ has access to the Principle of Induction, it can prove for each **FIXED** j , there exists some **FIXED** k such that $\forall x \forall y$ if x is a Godel number of the form $[\Phi(\hat{v}[1])]$ and $y = kx$ then $\exists z < y$ which is the Godel number of $[\nu\#[1, j] \Phi(\hat{v}[1])]$. (The verification of this fact is not hard.) Since the italicized statement represents a Π_1 theorem of PA^+ , the Group-2 axioms will imply that it is also a theorem of $IS(PA^+)$. Moreover, $IS(PA^+)$ recognizes scalar multiplication as a total function (since $kx = x+x+x+\dots+x$). The last two sentences imply that $IS(PA^+)$ can also prove that the mapping of $[\Phi(\hat{v}[1])]$ onto $[\nu\#[1, j] \Phi(\hat{v}[1])]$ is a total function. QED

Proof of Lemma 5.2's Second Sentence. Essentially the same as the preceding paragraph, with the roles of j and Φ reversed.

Added Comment: Proofs analogous to the above are used *repeatedly* in this paper. They will show how $IS(PA^+)$'s Group-2 Axioms, together with the fact that its Group-1 axioms can recognize scalar multiplication as a total function, identifies other functions which $IS(PA^+)$ can also recognize as total. We will return to this subject when Lemma 5.3 discusses the $u\#$ notation.

To encode the underlying tree structure of a semantic tableaux proof, we will need either pointers or parenthesis to formally represent the parent-child relationship. The terms **pointer encodings** and **parenthesis encodings** will refer to these two representations of a tree's underlying structure. Our results hold for either representation, and the parenthesis notation is preferable because it is more ideally compressed. Therefore $\{ P \{ \alpha \} \{ \beta \} \}$ denotes a tree whose root stores the byte string P , and where $\{ \alpha \}$ and $\{ \beta \}$ describe the possible subtrees descending from this root.

Notation. Recall that a **bounded quantifier** is defined to be an expression similar to " $\forall x < y$ " or " $\exists x < y$ ". Say an unbounded quantifier is **broad** iff its range, indicated by the symbols "(" and ")", includes all the atoms and bounded quantifiers in the sentences containing it. (Bounded quantifiers, by definition, will never be "broad".) Define a sentence to be in **Positive Normal Form** iff no broad quantifier has a $\neg$ symbols to its left. We will often desire to transform an initial sentence Ω into an alternate equivalent sentence written in positive normal form. The latter will be called the **Positive Rewrite** of Ω , and be denoted as **Positive(Ω)**. It can be defined as comprising that positive normal sentence which is identical to Ω except that:

- i) all of the " $\neg$ " symbols appearing to the left of Ω 's broad quantifiers are removed from **Positive(Ω)**
- ii) to preserve the logical equivalence of Ω and **Positive(Ω)** after operation (i) is performed, a " $\neg$ " symbol will be placed to the right of Ω 's rightmost broad quantifier (if necessary), and the " $\forall$ " and " $\exists$ " symbols in Ω 's and **Positive(Ω)**'s broad quantifiers are interchanged when necessary.

For example, Equation (5.2) is the positive rewrite of (5.1).

$$\neg \forall v \exists w \neg \forall x \neg \forall y (v=w \wedge \exists z=x+y) \quad (5.1)$$

$$\exists v \forall w \forall x \exists y \neg (v=w \wedge \exists z=x+y) \quad (5.2)$$

General definitions of semantic tableaux proofs can be found in [Fi90,Sm68]. The variant studied in this paper will define a semantic proof of Ψ as a tree of sentences where

- 1) The root consists of the sentence $\text{Positive}(\neg\Psi)$, (i.e. it represents the temporary contradiction assumption that Ψ is false). Each other node consists of an axiom or a deduction from a higher node in the tree using the standard rules [Fi90,Sm68] for eliminating the logical symbols $\wedge, \vee, \supset, \forall$ and $\exists$ and right-shifting the position of the symbol $\neg$. These rules are listed in Figure 1. The rule for eliminating existential quantifiers will require that the node inferring $\Phi(\hat{u}[i])$ from $\exists v[j] \Phi(v[j])$ be the child of the latter and that no ancestor of these two nodes will have previously introduced the parameter $\hat{u}[i]$.
- 2) Every tree branch is "closed" (by containing both Φ and $\neg\Phi$ for some sentence Φ).

Semantic tableaux proofs are known to satisfy the Godel Completeness Theorem [Fi90, Sm68]. Hence in some sense they have power commensurate to Hilbert-style proofs.

The definitions in this chapter were chosen carefully so that it would be easy to transform a proof of $\forall \hat{v}[j] \Phi(\hat{v}[j])$ into a proof of $\Phi(\hat{c}[k])$. Thus, if the initial axiom system is consistent and has its axiom sentences written in positive normal form, then two nodes in $\forall \hat{v}[j] \Phi(\hat{v}[j])$'s proof tree must be the root $\exists \hat{v}[j] \neg\Phi(\hat{v}[j])$ and its child $\neg\Phi(\hat{u}[i])$. A corresponding proof of $\Phi(\hat{c}[k])$ can be constructed by replacing these two nodes with $\neg\Phi(u\#[i,k])$. Since no other changes are necessary, this transformation will increase the proof's Godel number by no more than a *fixed constant multiplicative factor*. This fact will be employed to prove Lemma 5.3.

Lemma 5.3. Let p be the Godel number of a proof of $\forall \hat{v}[j] \Phi(\hat{v}[j])$ and $T(p,k)$ denote the transformed proof of $\Phi(\hat{c}[k])$, as defined above. For any **FIXED** k , the mapping of p onto $T(p,k)$ can be recognized by $\text{IS}(\text{PA}+)$ to be a total function. Likewise for **FIXED** p , $\text{IS}(\text{PA}+)$ can recognize the mapping of the byte string " $\hat{c}[k]$ " onto $T(p,k)$ to be a total function.

Proof. Both claims of Lemma 5.3 have proofs similar to Lemma 5.2. We will therefore sketch only the proof of the first claim. Since PA^+ has the strength of Peano Arithmetic, it can prove the mapping of the Godel number p onto $T(p,k)$ is total and satisfies the inequality $T(p,k) \leq L_k \cdot p$ (where L_k is a constant depending on k). Hence, PA^+ can prove the Π_1 sentence that $\forall p \forall x$ if $x = L_k \cdot p$ and p is a proof of $\forall \hat{v}[j]\Phi(\hat{v}[j])$ then $\exists q < x$ which is a proof of $\Phi(\hat{c}[k])$.

Since the italicized statement represents a Π_1 theorem of PA^+ , the Group-2 axioms will imply that it is also a theorem of $IS(PA^+)$. As before, $IS(PA^+)$ will recognize multiplication by any *scalar constant* as a total function because $L_k p = p+p+p+\dots+p$. The last two sentences imply that $IS(PA^+)$ can also prove that the mapping of p onto $T(p,k)$ to be a total function. QED

It should be noted that Lemma 5.3 would not follow, had this section employed slightly different definitions (such as omitting the symbol $u\#$), because $T(p,k) \leq L_k p$ could then fail for all constants L_k . Our definitions had guaranteed $T(p,k) \leq L_k p$ because, under normal human reasoning, the proof of $\Phi(\hat{c}[k])$ appears to never have much more complexity than the proof of $\forall \hat{v}[j]\Phi(\hat{v}[j])$. In any case, even without such a formalism supporting Lemma 5.3, most of our seven main theorems, including Propositions 1, 2a, 3, 5, & 7 will remain valid.

A formal 16-page description of the Godel encodings of $Prf_{IS(A)}(x,y)$ as a Δ_0 formula appears in Appendix A. In the remainder of this section, we will briefly summarize the salient aspects of $Prf_{IS(A)}(x,y)$'s encoding.

By applying Lemma 5.1, it is not hard to verify that:

- I) There exists a Δ_0 formula Turing (x, y, z) which states x is an encoding of the first y states of a Turing Machine whose initial state is z ;
- II) If an axiom system A has a Δ_0 formula $Ax_A(a)$ indicating that a is an axiom of A , then there will exist a Δ_0 formula $Prf_A(x, y)$ indicating that y is a semantic tableaux proof of x from A .

While it is not hard to apply Lemma 5.1 to prove I and II, it is substantially harder to verify that the hypothesis of II also implies that there exists Δ_0 formulae for representing $Prf_{IS(A)}(x,y)$, $Prf_{IS^{z_1}(A)}(x,y)$, $Prf_{IS^{u_{z_1}}(A)}(x,y)$, etc. The Godel encodings for the latter Δ_0 formulae is complex because it employs fix point operators to represent self verification. It is difficult to briefly summarize Appendix A's 16-page construction, but the proof of Lemma 5.4 (below) should explain at least some of the intuition.

Lemma 5.4. Let $\text{Prf2}_B(x, y_1, y_2)$ denote a Δ_0 formula indicating y_1 is a semantic tableaux proof of x from B , and y_2 is a description of the successive states of a Turing Machine that verifies y_1 is well defined. Define $\text{IS2}_d^f(A)$ to be the analog of $\text{IS}_d^f(A)$ that uses the Δ_0 formula $\text{Prf2}_{\text{IS}_d^f(A)}(x, y_1, y_2)$ in the place of $\text{Prf}_{\text{IS}_d^f(A)}(x, y)$. Then for all nice A , there are Δ_0 encodings for $\text{Prf2}_{\text{IS2}(A)}(x, y_1, y_2)$, $\text{Prf2}_{\text{IS2}^{\Sigma_1}(A)}(x, y_1, y_2)$, $\text{Prf2}_{\text{IS2}^{\text{U}\Sigma_1}(A)}(x, y_1, y_2)$, etc.

Quite Informal Proof Sketch. Kleene's Fixed Point Theorem [Ke38] is known [Ro67] to imply that every r.e. axiom system A can be extended into a r.e. system *which formally asserts* (albeit often incorrectly) its own consistency. Rogers points out that this result holds for all the standard definitions of self-consistency (thus for inability to prove $0=1$, Σ_1 local reflection, Σ_1 uniform reflection, etc.). Since the variable y_2 in $\text{Prf2}(x, y_1, y_2)$ essentially simulates r.e. computations, Kleene's Fixed Point Construction can be shown to imply that each of the axiom systems $\text{IS2}(A)$, $\text{IS2}^{\Sigma_1}(A)$, $\text{IS2}^{\text{U}\Sigma_1}(A)$ etc. can be provided with formal r.e. encodings. Since Turing (x, y, z) can be encoded as a Δ_0 formula, it then follows that each of $\text{Prf2}_{\text{IS2}(A)}(x, y_1, y_2)$, $\text{Prf2}_{\text{IS2}^{\Sigma_1}(A)}(x, y_1, y_2)$, $\text{Prf2}_{\text{IS2}^{\text{U}\Sigma_1}(A)}(x, y_1, y_2)$, etc. have corresponding Δ_0 definitions. QED.

The preceding proof sketch was deliberately kept "quite informal", and its only purpose was to illustrate intuitively how the encoding of $\text{Prf2}_{\text{IS}(A)}(x, y)$ is related to Kleene's Fixed Point Theorem. In Appendix A, the discussion is both more rigorous and stronger (in that it applies to the Prf rather than the Prf2 formula). We strongly recommend that the reader postpone examining Appendix A's formal Δ_0 encoding of $\text{Prf}_{\text{IS}(A)}(x, y)$ until after he has finished Section 6-14 because Appendix A is tedious and distracts from the more important themes of Sections 6-14. Moreover, Appendix A is in some sense unnecessary because each theorem in Sections 6-14, about IS, retains meaning, even without Appendix A, if Lemma 5.4's slightly weaker IS2 formalism is used to replace IS. That is, each theorem about IS can be translated into an analogous result about IS2 by simply replacing y with the ordered pair (y_1, y_2) .

6. Proof of Proposition 1

In this section, we will frequently refer to the subcomponents of a particular sentence. The definition of this construct is slightly different from the notion of a "subformula". Define a relation $\langle X, Y \rangle$ which specifies X is a subcomponent of Y to be the minimal relation satisfying the following five constraints:

- i) The formulae X and Y will be subcomponents of $X \vee Y$ and $X \wedge Y$;
- ii) The formulae $\neg X$ and Y will be subcomponents of $X \supset Y$;

iii) For any parameter u or constant c , the formulae $X(u)$ and $X(c)$ will be subcomponents of $\forall a X(a)$ and $\exists a X(a)$;

iv) X is a subcomponent of $\neg\neg X$; other rules for the symbol $\neg$ are $\langle \neg X \wedge \neg Y, \neg(X \vee Y) \rangle$, $\langle \neg X \vee \neg Y, \neg(X \wedge Y) \rangle$, $\langle X \wedge \neg Y, \neg(X \supset Y) \rangle$, $\langle \forall a \neg X(a), \neg \exists a X(a) \rangle$, and $\langle \exists a \neg X(a), \neg \forall a X(a) \rangle$.

v) If X is a subcomponent of Y and Y is a subcomponent of Z , then X will also be a subcomponent of Z .

Every semantic tableaux or resolution proof [Fi90, Sm68] of $\perp$ from an axiom system A must have every sentence of the proof tree (except possibly its root) constitute a subcomponent of some axiom of A . This characteristic of semantic tableaux and resolution proofs will be called their "cut-free" property. It is closely analogous to Gentzen's notion [Ge69, Ta87] of a cut-free sequent calculus proof. Whenever the subscript d from $IS_d^y(A, G)$ is omitted in this paper, our theorem technically applies to semantic tableaux deduction, although it trivially generalizes to resolution, as well as any formal "cut-free" deduction method.

Let $\text{Prf}_{IS(A)}(\perp, y)$ denote the provability formula from Equation (2.2), which asserts that y is a semantic tableaux proof of $\perp$. Then Equation (6.1) indicates that $IS(A)$ is inconsistent:

$$\neg \forall y \neg \text{Prf}_{IS(A)}(\perp, y) \quad (6.1)$$

Lemma 6.1 The sentence (6.1) is not a subcomponent of any Group 1 or 2 axiom of $IS(A)$.

Proof Sketch. It is clear that (6.1) is not a subcomponent of any Group-1 axiom. The canonical form for Group-2 axioms appeared in (2.1), and it is rewritten below:

$$\text{Pr}_A(\lceil \Phi \rceil) \supset \Phi \quad (2.1\text{-rewritten})$$

Since the definition of Group-2 requires that Φ be a Π_1 sentence, it is clear that sentence (6.1) can not be subcomponent of Φ . Also, the definition of $\text{Pr}_A(\lceil \Phi \rceil)$ implies that (6.1) can not possibly be a subcomponent of $\neg \text{Pr}_A(\lceil \Phi \rceil)$. QED

Lemma 6.2 If the axiom system A is nice then a semantic tableaux proof p of $\perp$ from $IS(A)$ is impossible without p formally constructing a parameter or constant element p^* , such that one node of p 's proof tree is the sentence $\neg \text{Prf}_{IS(A)}(\perp, p^*)$.

Proof A proof of $IS(A)$'s inconsistency would consist of a tree whose axioms come from $IS(A)$ and whose branches are all closed. Since A is nice, the Group 1 and 2 axioms must be

certainly consistent with each other and valid under the standard model of the natural numbers. Hence, an inconsistency proof p must use the Group 1 and 2 axioms to contradict the self-consistency statement (2.2), which is rewritten below:

$$\forall y \neg \text{Prf}_{\text{IS(A)}}(\perp, y) \quad (2.2\text{-rewritten})$$

But Lemma 6.1 showed assertion (6.1) was not a subcomponent of any of the Group-1 or Group-2 axioms! Hence, the necessary contradiction of (2.2) is technically possible only if the Group-1 and Group-2 axioms contradict a subcomponent of (2.2) of the form $\neg \text{Prf}_{\text{IS(A)}}(\perp, p^*)$, rather than actually contradict (2.2) directly. QED

We will now introduce some further notation that will simplify the main proof. Rather than use an m -ary function symbol $f(x_1, x_2, \dots, x_m)$ to denote a function f , we will represent each function with $(m+1)$ -ary relation symbol $A_f(x_1, x_2, \dots, x_m, w)$ together with the added axiom $\forall x_1 \forall x_2 \dots \forall x_m \exists w : A_f(x_1, x_2, \dots, x_m, w)$.

In this context, the semantic tableaux rule for eliminating an existential quantifier will replace $\exists w \Psi(w)$ with a sentence of the form $\Psi(u)$, where u is the symbol for a newly introduced parameter. Similarly, the rule for eliminating the universal quantifier in $\forall w \Phi(w)$ will replace it with the sentence $\Phi(k)$, where k represents here *either* one of the natural numbers $\bar{0}, \bar{1}, \bar{2}, \bar{3}, \dots$, or a parameter u that was introduced during the elimination of an existential quantifier at a tree node position which is an ancestor to the node specifying $\Phi(u)$.

In the rest of this section, we will assume that p denotes the minimal Godel number satisfying $\text{Prf}_{\text{IS(A)}}(\perp, p)$, and we will derive a contradiction by showing that a smaller p^* must also satisfy this formula. Let N denote a node in the proof tree p . Define $\text{ANCESTOR}(N)$ to be the list of all the sentences in the proof p that lie in the node N or one of its ancestors *but which exclude from this set any sentence of the exact form (2.2)* (i.e. excluded from $\text{ANCESTOR}(N)$ is the Group-3 axiom which formally asserts IS(A) 's self-consistency).

Say N is ancestor-consistent iff there exists an "interpretation" function, denoted as INT , that assigns an integer value to each parameter symbol u_i , such that all IS(A) 's Group-1 and 2 axioms and all the sentences in $\text{ANCESTOR}(N)$ are valid (under the Standard Model) when the symbols u_i are assigned their designated integer values under INT . Call the node N maximal iff it is a node of maximal depth in the proof tree p possessing some such interpretation function.

It is easy to apply the preceding definitions and lemmas to verify

- 1) that a maximal node N has an unique child N^* ,
- 2) that N^* stores a sentence of the form $\neg \text{Prf}_{\text{IS}(A)}(\perp, p^*)$ (where p^* represents either a constant or a parameter),
- 3) that N^* has no ancestor-consistent interpretation function (because N was maximal).

We will now derive a contradiction by showing that these three conditions force p^* to represent another inconsistency proof which is strictly less than p .

Let $\overline{m}_1, \overline{m}_2, \overline{m}_3 \dots$ designate the integer constants and $u_1, u_2, u_3 \dots u_j$ denote the parameter symbols appearing in the branch of p 's proof tree associated with N 's ancestors. Let $\overline{m}_{\max}$ denote the largest of $\overline{m}_1, \overline{m}_2, \overline{m}_3 \dots$. Then since the proof p consists of only $\log_2 p$ bits and since more than j of these bits are needed to represent $u_1 u_2 \dots u_j$, it must follow that

$$\overline{m}_{\max} < p \cdot 2^{-j} \quad (6.2)$$

We will now calculate the value of $\text{INT}(u_i)$ for each u_i . Without loss of generality, let us also assume that $\text{INT}(u_1) \leq \text{INT}(u_2) \leq \text{INT}(u_3)$ etc. We claim it must be true that

$$\text{INT}(u_j) \leq 2^j \cdot \overline{m}_{\max} \quad (6.3)$$

The proof of (6.3) rests on the facts that Addition is $\text{IS}(A)$'s only *increasing* function and that the existential quantifiers in the Group-2 axioms are *nonincreasing* (because of the requirement that Φ in a Group-2 axiom be a Π_1 sentence). Thus if u_i is the largest constant before the addition function defines a new parameter u_{i+1} then it must be true that $\text{INT}(u_{i+1}) \leq 2 \cdot \text{Max}(\text{INT}(u_i), \overline{m}_{\max})$. Hence, equation (6.3) follows because no more than j doublings are associated with $u_1 u_2 \dots u_j$, and the largest constant before these doublings occur is bounded by $\overline{m}_{\max}$.

The combination of (6.2) and (6.3) imply that $\text{INT}(p^*)$ must satisfy (6.4) (in both the cases where p^* represents a constant and a parameter).

$$\text{INT}(p^*) \leq \text{MAX}(\text{INT}(u_j), \overline{m}_{\max}) < p. \quad (6.4)$$

But INT becomes inconsistent on the tree branch associated with N^* at precisely the point where N^* introduces the sentence $\neg \text{Prf}_{\text{IS}(A)}(\perp, p^*)$. This implies that p^* must represent (under INT 's interpretation) an element smaller than p satisfying

$$\text{Prf}_{\text{IS}(A)}(\perp, p^*) \quad (6.5)$$

This is the desired contradiction, since p was presumed to be the smallest inconsistency proof. QED

7. Proof of Proposition 2

Define a terse proof of Ω to be a tree of sentences, partially analogous to a semantic tableaux proof, satisfying the following two rules:

- i) Each node in a terse proof will be either an axiom or a deduction from a higher node of the proof tree. (This property is the same as its analog in semantic tableaux proofs except that the root of Ω 's terse proof does not store the contradictory assumption $\neg \Omega$).
- ii) Terse proofs *will not* be required to satisfy the semantic tableaux condition that all branches of the proof tree be "closed" (by containing a pair of contradictory sentences). Instead, one (and only one) branch will be allowed to be an exception to this rule. The exceptional branch in Ω 's terse proof will be called its **pivotal branch**, and its leaf will instead contain the sentence $\text{Positive}(\Omega)$. (This sentence was defined in section 5 as essentially the rewriting of the sentence Ω in positive normal form).

Terse proofs are a much weaker mode of deduction than semantic tableaux proofs because Part (i) of their definition precluded performing proofs by contradiction. Their significance is easiest to appreciate if one considers the example where t represents a terse proof of $\neg \Phi$ and p represents a semantic tableaux proof of Φ . Let $\text{Glue}(t,p)$ denote the proof tree whose root is $\{\neg \perp\}$ (i.e. the sentence $\{0 \neq 1\}$), which has the subtree t descend from the root, and which has the proof tree p descend from the bottom of t 's pivotal branch. It is easy to see that $\text{Glue}(t,p)$ is a semantic tableaux proof of $\perp$.

But since it is impossible for a consistent axiom system to have a semantic tableaux proof of $\perp$, it follows that *no consistent axiom system can simultaneously have a terse proof of $\neg \Phi$ and a semantic tableaux proof of Φ* . The non-trivial aspect of the italicized sentence is that it is *not merely a meta-theorem*: Rather Lemma 7.1 will show that the axiom system $\text{IS}(\text{PA}+)$ will have sufficient power to formally prove an essentially self-reflexive form of this sentence!

While examining this chapter, the reader should bear in mind that our goal is to study semantic tableaux proofs (rather than the weaker terse proofs), and to formally prove Proposition 2. The terse proofs in Lemma 7.1 will serve solely as a vehicle for analyzing the properties of semantic tableaux proofs, and for proving Proposition 2 later in this chapter.

Lemma 7.1 If there exists a *terse* proof tree for $\neg \Phi$ employing the axioms from $\text{IS}(\text{PA}+)$, then $\text{IS}(\text{PA}+)$ *can prove* that it will be *unable* to construct a *semantic tableaux* proof of Φ .

Proof. The hypotheses of Lemma 7.1 indicates that there exists a constant g representing the Gödel number of $\neg\Phi$'s terse proof. Let $k = 64 \cdot g \cdot \lceil \neg\perp \rceil$ denote a second constant (which must therefore also exist). While $\text{IS}(\text{PA}^+)$ can not recognize multiplication as a function, it certainly can verify the scalar product $kp = p+p+p+\dots+p$ is a function. Thus it can prove

$$\forall p \exists y \{ y = kp = 64 \cdot \lceil \neg\perp \rceil \cdot g \cdot p \} \quad (7.1)$$

Since the axiom system PA^+ is stronger than Peano Arithmetic, it can verify *neither* its consistency *nor* the consistency of $\text{IS}(\text{PA}^+)$. However, PA^+ is strong enough to verify that “ $\forall t \forall p \forall y$ if t is a terse proof of $\neg\Phi$ and p is a semantic tableaux proof of Φ and $y = 64 \cdot \lceil \neg\perp \rceil \cdot t \cdot p$ then $\exists q < y$ satisfying $\text{Prf}_{\text{IS}(\text{PA}^+)}(\perp, q)$ ”. (In particular, any system as strong as Peano Arithmetic can prove that $\text{Glue}(t, p)$ satisfies the properties required of q .)

Since the bold-face sentence is a Π_1 theorem of PA^+ , the Group-2 axioms imply it is also a theorem of $\text{IS}(\text{PA}^+)$. Moreover, since $\text{IS}(\text{PA}^+)$ can prove (7.1) for some g which is a terse proof, it can also recognize that the preceding bold face statement implies “ $\forall p$ if p is a semantic tableaux proof of Φ then $\exists q$ satisfying $\text{Prf}_{\text{IS}(\text{PA}^+)}(\perp, q)$ ”. Lastly, $\text{IS}(\text{PA}^+)$ can recognize that the second statement in bold print, combined with the Group-3 axiom “ $\forall q \neg \text{Prf}_{\text{IS}(\text{PA}^+)}(\perp, q)$ ”, implies “ $\forall p \neg \text{Prf}_{\text{IS}(\text{PA}^+)}(\lceil \Phi \rceil, p)$ ”. Hence, $\text{IS}(\text{PA}^+)$ can prove “ $\forall p \neg \text{Prf}_{\text{IS}(\text{PA}^+)}(\lceil \Phi \rceil, p)$ ” (see the the next paragraph for an important clarifying comment). QED

Remark 2. In the preceding paragraph and often elsewhere in this paper, our proofs shall *apply a cut rule at the meta-theoretical level*. That is, suppose it is known that a *cut-free* $\text{IS}(\text{PA}^+)$ system can prove the two statements α and $\alpha \supset \beta$. Then by the Cut Elimination Theorem [Be56, Ge69, Sc56, Fi90, Sm68, Ta87], $\text{IS}(\text{PA}^+)$ will also support a cut-free proof of β . For the sake of shortening many of the proofs in this paper, we will *often omit constructing formally the cut-free proofs* of such theorems β : Instead, *their theoretical existence* will be established implicitly via the Cut Elimination Theorem and the fact that proofs for α and $\alpha \supset \beta$ *exist!* Such meta-reasoning had greatly simplified the proof of Lemma 7.1. In particular, if α and β designate the two bold-face sentences in Lemma 7.1's proof, then our proof had directly implied only that $\text{IS}(\text{PA}^+)$ could prove α and $\alpha \supset \beta$. Meta-Logic had then implicitly implied that $\text{IS}(\text{PA}^+)$ could also prove β ! Also, if γ and ω denote the sentences $\forall q \neg \text{Prf}_{\text{IS}(\text{PA}^+)}(\perp, q)$ and $\forall p \neg \text{Prf}_{\text{IS}(\text{PA}^+)}(\lceil \Phi \rceil, p)$, then the final line of Lemma 7.1's proof had used Meta-Logic and the fact that $\text{IS}(\text{PA}^+)$ could prove β , γ and $\beta \wedge \gamma \supset \omega$ to conclude that $\text{IS}(\text{PA}^+)$ could also prove ω itself. Meta-Logic will be used frequently in the remainder of this paper to shorten and simplify the proofs of many of our theorems.

We will now use Lemma 7.1 to prove Proposition 2a. *Consider first the case where Φ is false.* Since it is a false Δ_0 sentence, there must exist a constant c satisfying $\text{Prf}_{\text{PA}+}(\ulcorner \neg\Phi \urcorner, c)$. Figure 2 illustrates how $\neg\Phi$ must then have a *terse proof* from the system $\text{IS}(\text{PA}+)$ (because the facts that c exists and that one of $\text{IS}(\text{PA}+)$'s Group-2 axioms is the sentence $\text{Pr}_{\text{PA}+}(\ulcorner \neg\Phi \urcorner) \supset \neg\Phi$ leads to a natural construction of a terse proof). Hence by Lemma 7.1, $\text{IS}(\text{PA}+)$ *can formally prove* there exists no proof of Φ . This implies that $\text{IS}(\text{PA}+)$ can formally verify Φ 's local reflection property when Φ is false. Proposition 2a is also valid when Φ is true because every true Δ_0 sentence is automatically proveable (thereby implying $\text{IS}(\text{PA}+)$ can trivially verify its reflection principle). QED

We will now use *meta-logic* (see Remark 2) and Lemmas 7.1 and 5.3 to prove Proposition 2b. For simplicity, assume the Π_1 sentence has the form $\Psi \equiv \{\forall x \Phi(x)\}$. *Consider first the case where Ψ is refutable.* Then since $\neg\Psi$ is Σ_1 , both $\text{PA}+$ and $\text{IS}(\text{PA}+)$ can prove $\neg\Phi_{\bar{k}}$ for some integer $\bar{k}$ (see footnote *). Since $\neg\Phi_{\bar{k}}$ is Δ_0 , we can again construct a terse proof of it. The new facet is that we can now use Lemma 5.3 to map every proof q of $\forall x \Phi(x)$ onto a proof $f_{\bar{k}}(q)$ of $\Phi_{\bar{k}}$, where $f_{\bar{k}}$ is a *provable function under $\text{IS}(\text{PA}+)$* . But Lemma 7.1 in the context of $\neg\Phi_{\bar{k}}$'s terse proof implies $\text{IS}(\text{PA}+)$ *can prove it has no semantic tableaux proof of $\Phi_{\bar{k}}$* . Hence since $\text{IS}(\text{PA}+)$ *can verify $f_{\bar{k}}$ is a function*, it must be able to confirm that there is no proof of $\forall x \Phi(x)$. This shows $\text{IS}(\text{PA}+)$ can verify the local reflection statement for $\forall x \Phi(x)$ when this sentence is refutable.

The proof for the reflection statement of $\forall x \Phi(x)$ is trivial in the alternate case where this sentence is verifiable. Hence the reflection statement for $\forall x \Phi(x)$ is provable in $\text{IS}(\text{PA}+)$ whenever this sentence is decidable. QED

A similar methodology is used to prove Proposition 2c. Proposition 2a indicates that $\text{IS}(\text{PA}+)$ can prove (7.2) when Φ_i is a Δ_0 sentence and i is fixed.

$$\{\exists y \text{Prf}_{\text{IS}(\text{PA}+)}(\ulcorner \Phi_i \urcorner, y)\} \supset \Phi_i \quad (7.2)$$

Also Lemma 5.3 indicates that $\text{IS}(\text{PA}+)$ can certainly prove

$$\{\exists x \text{Prf}_{\text{IS}(\text{PA}+)}(\ulcorner \forall v \Phi(v) \urcorner, x)\} \supset \{\exists y \text{Prf}_{\text{IS}(\text{PA}+)}(\ulcorner \Phi_i \urcorner, y)\} \quad (7.3)$$

Since $\text{IS}(\text{PA}+)$ can prove (7.2) & (7.3), it must be able to prove also (7.4), by the transitive version of the Cut Elimination Theorem (defined below).

$$\{\exists x \text{Prf}_{\text{IS}(\text{PA}+)}(\ulcorner \forall v \Phi(v) \urcorner, x)\} \supset \Phi_i \quad \text{QED} \quad (7.4)$$

Definition. The transitive version of the Cut Elimination Theorem is used often in this paper. It indicates that if $\text{IS}(\text{PA}+)$ can prove $\gamma \supset \omega$ and $\omega \supset \mu$, then it can also prove $\gamma \supset \mu$.

* From ZF set theory, one can infer that such a $\bar{k}$ must exist because Proposition 1 implies that the standard model for the Natural Numbers is a model for $\text{IS}(\text{PA}+)$.